higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ENGINEERING SCIENCE N4

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This marking guideline consists of 12 pages.

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SECTION A

QUESTION 1

1.1 In a frictionless open environment, when a ball is hit by a baseball and the ball is moving at 20 m/s, it will keep on moving at 20 m/s for ever, unless it hits an object on its path and its velocity is reduced or its direction changes.

OR

When an object is stationary on a table, it will never change its position unless something moves it.

OR

A marble in motion comes to rest because of the friction which opposes its motion/If the friction is completely eliminated, the marble will preserve its state of motion.

OR

An object thrown vertically upwards gets retarded and momentarily comes to rest at a certain height above the ground due to the force of gravity which opposes its motion. If the gravity is eliminated, the object preserves its state of uniform motion. (Any 1 × 2) (2)

1.2 1.2.1 Angular acceleration is the rate of angular velocity and it is measured in rad/s.

1.2.2 Stress is the reaction on a material due to an external force.

(2 × 1) (2)

1.3 1.3.1 $\beta = 2\alpha$
$\gamma = 3\alpha$ (2)

1.3.2 $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ (1)

1.4 Hooke's law states that at a constant temperature, the pressure of a gas is inversely proportional to its volume. (2)

[9]

QUESTION 2

2.1	D		(1)
2.2	2.2.1	A	(2)
	2.2.2	A	(2)
2.3	A		(1)
2.4	D		(1)
2.5	C		(1)
2.6	D		(1)
2.7	D		(2)
2.8	C		(2)
2.9	B		(2)
			[15]

TOTAL SECTION A: 24

SECTION B

QUESTION 3

3.1 S_h = 55 km
U = 520 km/h

$\theta = 50^o$

$u = 520 \ \ km/h$

=144,444 m/s✓

$\theta = 50°$

$S_h = u.t\cos\theta$

$t = \dfrac{S_h}{u.\cos\theta}$

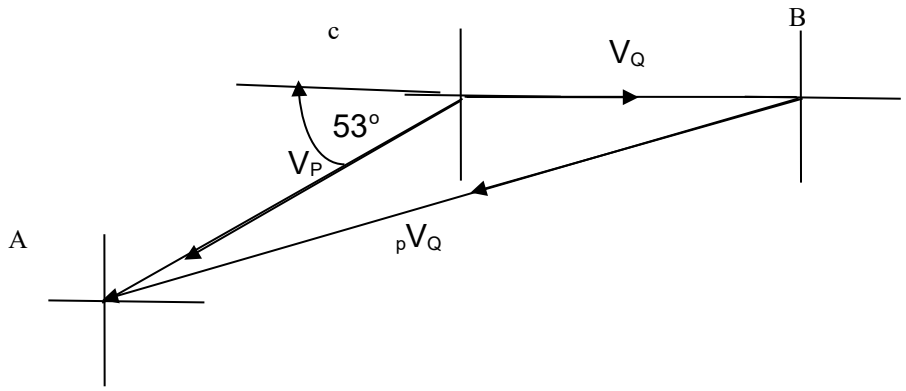
$\dfrac{55\ \ 000}{144{,}444.\cos 50}$ ✓

$= 592{,}375 \ \ s$ ✓

$= 9{,}873 \ \ minutes$ ✓ (4)

3.2 $V_p = 250 \ \ km/h$ E

$V_Q = 210 \ \ km/h \ \ W\,53^o\,S$



$C = 37° + 90°$

$= 127°$

$c = \sqrt{210^2 + 250^2 - 2(210).(250)\cos 127}$ ✓

$= 412{,}057 \ \ km/h$ ✓

$\dfrac{\sin B}{b} = \dfrac{\sin C}{c}$

$B = \sin^{-1}\dfrac{b.\sin C}{c}$

$\sin^{-1}\dfrac{(210).\sin 127°}{412{,}057}$ ✓

$= 24{,}018°$ ✓

✓

$_pV_Q = 412{,}057 \ \ km/h \ \ W\,24{,}018°\,S$ (5)

[9]

QUESTION 4

4.1 V 280 km/h
d = 720 mm

$v = \pi Dn$

$n = \frac{v}{\pi D}$

$n = \frac{77,7778 \ m/s}{\pi(0,72)}$

$= 34,385$✓

N= 2 063,12 r/min✓

ALTERNATIVE

$v = \omega R$

$\omega = \frac{v}{R}$

$\omega = \frac{77,7778 \ m/s}{0,36 \ m}$

$= 216,0493833 \ rad/s$✓

$N = \frac{216,0493}{2\pi}$

N = 2 063,12 r/min✓ (2)

4.2 4.2.1 $N_1 = 444 \ r/min$

$N_2 = 2 \ 840 \ r/min$

$t = 0,58 \ min \omega_1 = 444 \ r/min \frac{2\pi}{60}$

$= 46, \ 49558 \ rad/s$✓

$\omega_2 = 2 \ 840 \ r/min \frac{2\pi}{60}$

$= 297, \ 40410445 \ rad/s$✓

$\omega_2 = \omega_1 + \alpha . t$

$\alpha = \frac{\omega_2 - \omega_1}{t}$

$= \frac{297,40410 - 47,9558}{34,8}$

$= 7,\ 210\ \ rad/s^2$ ✓

4.2.2 n

$\omega_2{}^2 = \omega_1{}^2 + 2\alpha\theta$

$\theta = \dfrac{\omega_2{}^2 - \omega_1{}^2}{2\alpha}$

$\theta = \dfrac{297{,}4041^2 - 46{,}49558^2}{2(7{,}21)}$ ✓

$= 17{,}400\ \ rad$ ✓

$n = \dfrac{\theta}{2\pi}$

$n = \dfrac{\theta\, 17{,}400}{2\pi}$

$= 2{,}769$ ✓

(2 × 3) (6)

[8]

QUESTION 5

5.1 m = 980 kg
Slope 1:45
u = 6 m/s
v = 21 m/s
f_u = 0,459 N/kg
t = 35 s

$f_u = 0{,}459\ \ N/\ kg$

A

$v = u + a.t$

$a = \dfrac{v - u}{t}$

$= \dfrac{21 - 6}{35}$ ✓

$= 0{,}42857\ \ m/s^2$ ✓ (2)

5.2 $f_a = ma$

$= 980(0{,}42857)$ ✓

$= 420\ \ N$ ✓ (2)

5.3 $f_a = 420\ \ N$

$f_u = 0{,}459(980)$

$= 449{,}82\ \ N$ ✓

$\theta = \sin^{-1}\left(\dfrac{1}{45}\right)$

$= 1{,}2739°$

$f_p = 980(9{,}8)\sin 1{,}2739°$

$= 213,4222 \ N$ ✓

$f_{tot} = f_p + f_u + f_a$

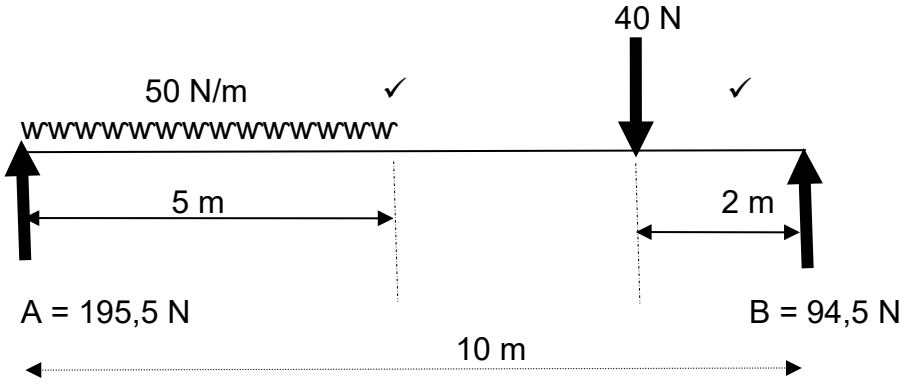
$f_{tot} = 213,4222 \ N + 449,82 \ N + 420 \ N$ ✓

$= 1 \ 083,242 \ N$ ✓ (4)

[8]

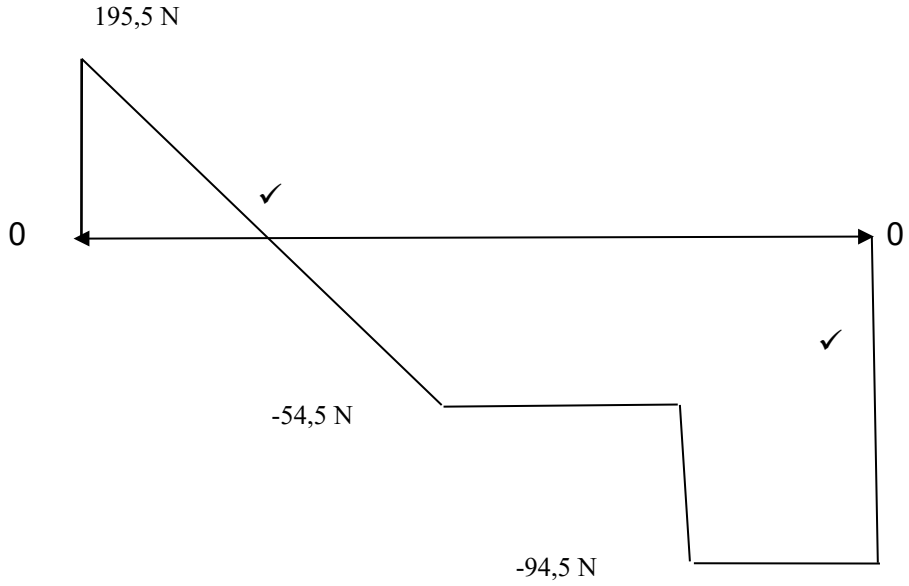
QUESTION 6

6.1 6.1.1



(2)

6.1.2



(2)

6.1.3 $BM_C \approx 195,5(5) - 50(5).\frac{5}{2} = 350 \ Nm$ ✓

$BM_D \approx 195,5(8) - 50(5)(5,5) - 40(0) = 189 \ Mm$ ✓

$\sum F._E = 0$

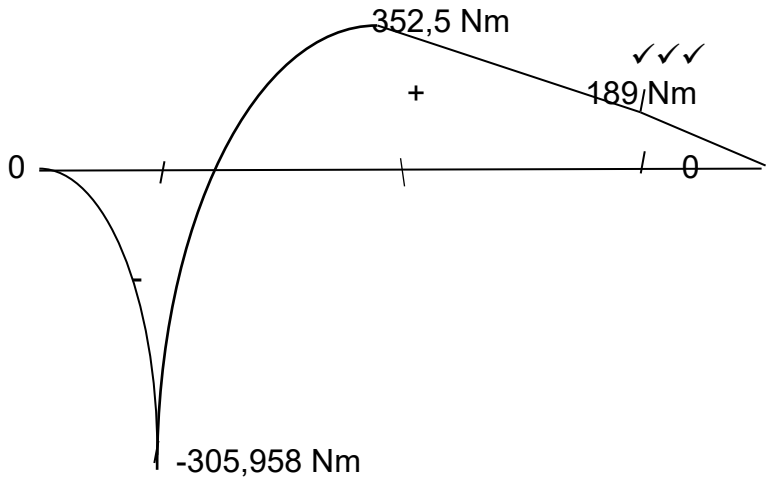
$195,5 \ N - 50 \ N/m. \ x = 0$

$x = \frac{-195,5 \ N}{-50 \ N/m}$

$= 3,91 \ m$ ✓ (3)

$$BM_E \approx 195,5(3,91) - 50(3,91).\frac{3,91}{2} = -305,958 \ \ Nm\checkmark$$

6.1.4



(3)

6.2

Shape	Area (cm^2)	Centroid (cm)	AC (cm^3)
Rectangle	L(b) = 60(50) = 3 000√	60/2 = 30√	3 000(30) = 90 000√
Triangular hole	0,5(12)18 = 108√	23 + 12/3 = 27√	108(27) = 2 916√
	$\sum A_1 - \sum A_2$ 3 000 -108 2 892√		$\sum AC_1 - \sum AC_2$ 90 000 – 2 916 = 87 084√
		$y = \frac{87\ 084}{2\ 892}$ $= 30,112 \ \ cm\checkmark$	

(5)

[15]

TOTAL SECTION B: 40

SECTION C

QUESTION 7

7.1 D = 50 m
h = 28 m
P = 250 kPa

7.1.1 $$V = \pi \frac{D^2}{4}.h$$
$$= \pi \frac{50^2}{4}.28$$
$$= 54\ 977{,}87144\ m^3 \checkmark$$

W = (250 000) 54 977,87144 m^3✓

$$= 1{,}374446786 \times 10^{10} J$$

$$= 13\ 744{,}468\ MJ \checkmark$$ (3)

7.1.2 P = 840 kW

$$P = \frac{W}{t}$$
$$t = \frac{W}{P}$$
$$t = \frac{13\ 744{,}46787 \times 10^6\ W}{840\ 000\ Pa} \checkmark$$

$$= 16\ 362{,}46174\ s$$
$$= 4{,}545\ hours \checkmark$$ (2)

7.2 3d = D
Stroke length = 100 mm
MA = 22
F_h, m = 2,2 Mg $\eta = 91\%$

$$W_R = (2\ 200).(9{,}8)$$
= 21 560 N✓

$$MA = \frac{f_p}{f_h}.\frac{100}{\eta} \qquad \frac{W_R}{f_p} = \frac{D^2}{d^2}$$

$$f_p = \frac{W_R d^2}{D^2}$$

$$f_p = \frac{21\ 560(d^2)}{(3d)^2} \checkmark$$

$$= 2\ 395.556\ N \checkmark$$

$$f_h = \frac{f_p}{MA} . \frac{100}{\eta}$$

$$f_h = \frac{2\ 395.556}{22} . \frac{100}{91} \checkmark$$

$$= 119{,}658\ N \checkmark$$ (5)

7.3 C = 2
d = 70 mm
Stroke length = 150 mm
$\eta = 94\%$
$V_a = 0{,}027\ l/s$

$$V_s = V_a . \frac{100}{\eta}$$

$$V_s = 0{,}027 . \frac{100}{94}$$

$$= 0{,}028723404\ l/s \checkmark$$

$$V_s = \frac{\pi(d^2) S_l . n}{4}$$

$$\frac{\pi(d^2) S_l . n . c}{4} \frac{N}{60} = 0{,}028723404\ l/s$$

$$\frac{\pi(0.07^2).(0.15).1.2.}{4} \frac{N}{60} = 0{,}028723404\ l/s \checkmark$$

$$N = \frac{0{,}028723404}{0{,}000019242}$$

= 1 492,725 r/s✓

= 24,879 r/min✓ (4)
[14]

QUESTION 8

8.1 d = 32 mm
l = 820 mm
F = 52,332 kN
$\Delta l = 0{,}621\ mm$

8.1.1 $\delta = \frac{f}{a}$

$a = \frac{\pi d^2}{4}$

$¿ \frac{\pi 0{,}032^2}{4}$

$= 0{,}000804247 \ m^2$✓

$\delta = \frac{52\ 332\ J}{0{,}000804247}$✓

$= 65{,}070 \ MPa$✓ (3)

8.1.2 $\in = \frac{\Delta l}{l}$

$¿ \frac{0{,}621}{820}$✓

$= 7{,}57317 \times 10^{-4}$

$= 7{,}573 \times 10^{-4}$✓ (2)

8.1.3 $E = \frac{\delta}{\varepsilon}$

$¿ \frac{65{,}070 \times 10^6}{7{,}57317 \times 10^{-4}}$ ✓

$= 8{,}592174 \times 10^{10}$

$= 85{,}922 \ GPa$✓ (2)

8.2 m = 1,8 tons
d = 5 mm
l = 50 m
$\Delta l = 8 \ mm$

$a = \frac{\pi d^2}{4}$

$¿ \frac{\pi 0{,}05^2}{4}$

$= 0{,}001963495 \ m^2$✓

$\delta = \frac{517\ 640\ N}{0{,}001963495\ m^2}$

$= 8\ 983\ 978{,}228 \ Pa$✓

$$\in = \frac{\Delta l}{l}$$

$$¿ \frac{0,08\ \ mm}{50}$$

$$= 0,0016$$

$$E = \frac{\delta}{\varepsilon}$$

$$¿ \frac{8\ \ 983\ \ 978,228\ \ Pa}{0,0016}$$

$$= 5\ \ 614\ \ 986\ \ 393\ \ Pa$$

$$= 5,615\ \ GPa$$ ✓

ALTERNATIVE

$$E = \frac{f\,.\,L}{a\,.\,\Delta l}$$

$$E = \frac{17\ \ 640\ \ N(50)}{0,001963495.(0,08)}$$ ✓✓

$$= 5,615\ \ GPa$$ ✓ (3)

[10]

QUESTION 9

9.1 Dimensions: 500 mm × 200 mm × 600 mm
t = 20 °C
$\Delta t =$ 245 °C
$a =$ 12,5 × 10^{-6}/K

9.1.1 Δl

$$\Delta l = l_o \alpha \Delta T$$
$$= 0,6\left(12,5 \times 10^{-6}\right)518\ \ K$$ ✓
$$= 0,003885\ \ m$$ ✓ (2)

9.1.2 $\Delta V = 3\,.\,V_o\,.\,\propto \Delta t$

$$V_o = 0,5(0,2).(0,6)$$ ✓

$$\Delta V = 3\,.\,V_o\,.\,\propto \Delta t$$
$$= 3(0,06).\left(12,5 \times 10^{-6}\right)518\ \ K$$ ✓
$$= 0,0011655\ \ m^3$$
$$= 1,167 \times 10^{-3}\ \ m^3$$ ✓ (3)

9.2 V_1 = 0,208 m^3 T1 = 293 °K
t_1 = 20 °C t_2 = 2 °C
P1 = 1 850 kPa V_2 =

9.2.1 V_2

$$\frac{V_1}{T_1}=\frac{V_2}{T_2}$$

$$V_2=\frac{V_1.T_2}{T_1}$$

$$¿\frac{0,208(275)}{293}\checkmark$$

$$=0,195\ \ m^3\checkmark$$

(2)

9.2.2 P_2

$$P_2=\frac{P_1.V_1.T_2}{T_1.V_2}$$

$$¿\frac{185(0,208)(275)}{293(0,089)}\checkmark\checkmark$$

$$=405,798\ \ kPa\checkmark$$

(3)

9.2.3
- Combination of Boyle's and Charles' laws
- There is no constant term/s

(2)

[12]

TOTAL SECTION C: 36
GRAND TOTAL: 100